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BYZSO review, 2026-09-13 —same content, six layouts

Template	Pages EN/ZH	What it is
neurips	26/25	NeurIPS 2023: letter, 5.5x9in block, Times 10/11, bold headings, 4pt/1pt title rules, bold-lead hanging captions
iclr	26/25	ICLR 2025: same block, small-caps title/headings/Abstract, parskip 0.5pc
orchestra	26/25	PaperOrchestra conventions: plain captions (label + period), figures at 0.95 linewidth, clearpage before references
arxiv	24/22	arXiv preprint (arxiv.sty look): 6.0in block, left-aligned title with rule, sans bold headings, indented paragraphs
springer	35/–	Springer LNCS-like: 12.2x19.3cm block on A4, run-in headings, indented paragraphs
journal	23/21	Single-column journal (ACS/RSC-like): 6.5in (EN) / 16cm A4 (ZH) block, sans bold numbered headings, tight leading

Each template below is shown with: page 1 (title + abstract), the Figure 1 page, and the Table 1 page. Springer pushes the wide Table 1 to page 34 (narrow column), so its table page is that one.

Synthesis Conditions of $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ and Its Structural Relatives: Evidence Grading, Route Matrix, and Transferability Limits

Anonymous Review Working Group

Abstract

The non-centrosymmetric silicate $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ has been grown from a high-temperature solution and its single-crystal structure solved, but direct evidence for its starting composition, its bulk phase region, and any independent reproduction is still missing. Here we grade the available evidence and compare the published compositions, thermal histories, containers, cooling programs, and characterization results for the target phase, the Zn-free tetragonal Ba–Y silicate family, structurally related Ba–Zn–Si–O compounds, and Y–Si–O process-reference systems. The existing experiments show that phase formation near the target is governed simultaneously by the Ba/Y/Si ratio, the Zn–Y–oxygen stoichiometry, the flux proportion, and the cooling path, and that single-crystal structural evidence is no substitute for bulk phase-purity and measured-composition analysis. On this basis we propose four priorities: establishing a local phase diagram around $\text{Ba}_5\text{Y}_{12}\text{ZnSi}_8\text{O}_{40}$, testing the composition–structure relationship along $\text{Ba}_5\text{Y}_{13-x}\text{Zn}_x[\text{SiO}_4]_8\text{O}_{8.5-x/2}$, running solid-state reaction and high-temperature solution growth in parallel, and using Mg/Co analogs to isolate the effect of ionic-size differences. A local two-step inorganic retrosynthesis model recommends ZnO, MgO, and Co_3O_4 , respectively, together with BaCO_3 , Y_2O_3 , and SiO_2 , as the preferred starting-material sets for the Zn, Mg, and Co targets; these are precursor hypotheses awaiting verification, nothing more. The experiments most likely to yield discovery are not further trials at isolated temperatures but a closed feedback loop that ties together composition grids, phase quantification, single-crystal structure, measured stoichiometry, and process records.

1 Introduction

This review concerns the non-centrosymmetric barium yttrium zinc silicate $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$. We pursue three questions: whether the composition and structural assignment of the target phase can be confirmed; under which phase-formation conditions the phase can be obtained reproducibly; and whether a continuous composition–structure relationship connects the Zn-containing phase to the Zn-free tetragonal Ba–Y silicates. In the original report, crystals were grown from a high-temperature solution in an open system and the structure was solved by single-crystal X-ray diffraction (SCXRD) [1]. The structure combines a framework of highly coordinated rare-earth–oxygen polyhedra with isolated or composite silicate units, and it offers a reference point both for how complex silicates assemble and for the compositional design of non-centrosymmetric oxides.

The direct report of the target phase establishes only that the crystals exist and what their structure is; it gives no independently reproducible starting composition, no phase-formation range, and no route to phase-pure bulk samples [1]. We therefore do not stitch together a “best recipe” from the conditions used for structurally related compounds. Instead, we separate the questions that single-crystal growth, solid-state reaction, and phase-diagram screening can each answer, and we use that separation to

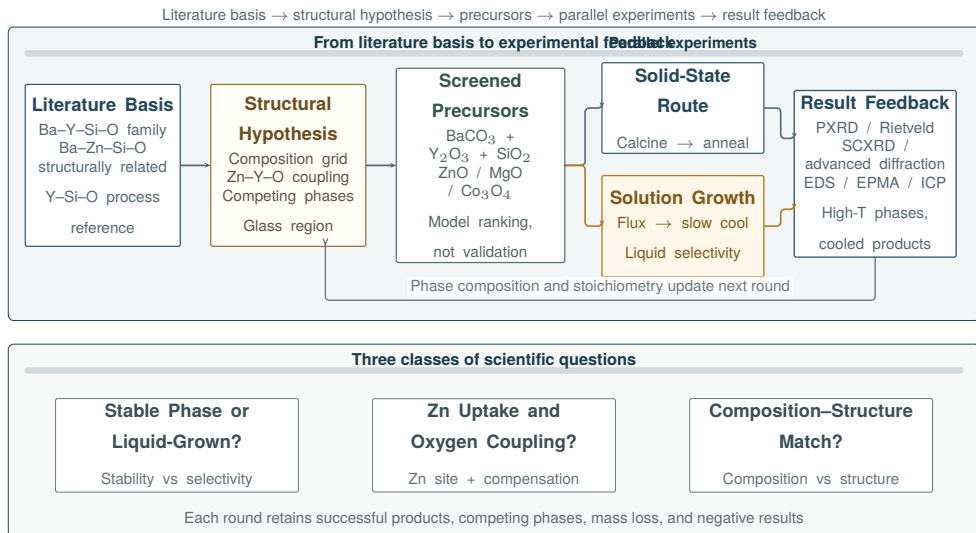


Figure 1: Research roadmap. The Ba-Y-Si-O, Ba-Zn-Si-O, and Y-Si-O systems are used first to establish structural relationships, competing phases, and process boundaries; phase-diagram and charge-balance analysis proposes candidate compositions, and the two-step inorganic retrosynthesis model supplies the precursor set; solid-state reaction and high-temperature solution growth probe the bulk phase region and the single-crystal structure, respectively, and the characterization results feed back into the next round of compositions and processes.

2 Literature Search and Evidence Assessment

2.1 Search Scope

The search began from the target formula in both its structural and its elemental spelling, using Ba₅Y₁₂Zn[O(SiO₄)]₁₈, Ba₅Y₁₂ZnSi₈O₄₀, BYZSO, and barium yttrium zinc silicate, combined with the keywords space group, crystal structure, flux, solid-state, phase diagram, and crystal growth. We then followed the literature in three directions: (i) the nomenclature and structural family of the Zn-free tetragonal Ba-Y silicates; (ii) Ba-Zn-Si-O compounds with firm structural evidence; and (iii) Y-Si-O systems for which high-temperature solution growth, solid-state reaction, mechanical activation, or Czochralski processes have been reported. A local full-text materials library was queried first; metadata and experimental passages were then verified through OpenAlex, Crossref, publisher pages, DOI resolution pages, IUCr, COD, RSC main texts and supplementary information (SI), and institutional repositories. The search cutoff was 2026-09-01. A miss in a database or interface means only that that channel returned nothing; it is not evidence that the literature does not exist.

2.2 Inclusion, Exclusion, and Deduplication

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Table 1: Compositional and structural evidence for the target compound and related Ba–Y silicates.

Records are grouped by how the subject of each publication relates to the target phase; neither the Zn-free studies nor the structurally related studies constitute an independent reproduction of the Zn-containing target phase.

Formula (original → as typeset here)	Ref.	Main structural evidence	Relation to target phase	Established and open
$\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$	[1]	Single-crystal X-ray structure determination [1]	Direct report of the target compound	Established: Zn-containing target phase and its route type; open: independent reproduction, bulk phase-pure window, correspondence to the Zn-free family
$\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ (tentative; x kept variable here)	[25, 42]	Reported as an accompanying phase; systematic phase-formation screening [25, 42]	Same-family tetragonal Ba–Y silicate	Established: historical name and composition-screening results; open: exact stoichiometry, structural model, relation to the target phase
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ (non-stoichiometric formula kept here)	[43]	Single-crystal revision to a non-stoichiometric modulated structure [43]	Same-family tetragonal Ba–Y silicate	Established: current structural reading of the historical formula; open: whether Zn can enter this family
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ ceramics	[44]	Bulk phase composition and microstructure [44]	Same-family tetragonal Ba–Y silicate	Established: bulk characterization requirements for the non-stoichiometric family; open: separating secondary phases and extraneous SiO_2 from the intrinsic phase region
$\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$	[18]	EDS, cRED, synchrotron PXRD, and neutron refinement [18]	Same-family tetragonal Ba–Y silicate	Established: independently confirmed tetragonal Ba–Y structure and phase-region reference; open: site correspondence to the Zn-containing target phase
$\text{BaY}_{16}\text{Si}_4\text{O}_{33}$	[30]	Single-crystal structure with isolated orthosilicate units [30]	Structurally related compound	Established: structural reference for Ba–Y orthosilicates; open: no evidence that it is the same phase as the Zn-containing target

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3.3 Reassessment of the Tetragonal Silicate Family

That a historical name is still in use does not mean the structure has carried over. An early flux-growth study recorded $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ as a tentative accompanying phase [25], and a later systematic flux study folded it into a multivariable phase-formation screen, but the subject remained the Zn-free tetragonal Ba–Y silicate [42]. Modern single-crystal work then recast the related nominal composition as the non-stoichiometric modulated structure $\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ [43], and the companion ceramic study showed that bulk samples must also be checked for secondary phases and for SiO_2 picked up from the grinding medium [44]. Historical names therefore have to be re-verified against

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1 INTRODUCTION

This review concerns the non-centrosymmetric barium yttrium zinc silicate $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$. We pursue three questions: whether the composition and structural assignment of the target phase can be confirmed; under which phase-formation conditions the phase can be obtained reproducibly; and whether a continuous composition–structure relationship connects the Zn-containing phase to the Zn-free tetragonal Ba–Y silicates. In the original report, crystals were grown from a high-temperature solution in an open system and the structure was solved by single-crystal X-ray diffraction (SCXRD) [1]. The structure combines a framework of highly coordinated rare-earth–oxygen polyhedra with isolated or composite silicate units, and it offers a reference point both for how complex silicates assemble and for the compositional design of non-centrosymmetric oxides.

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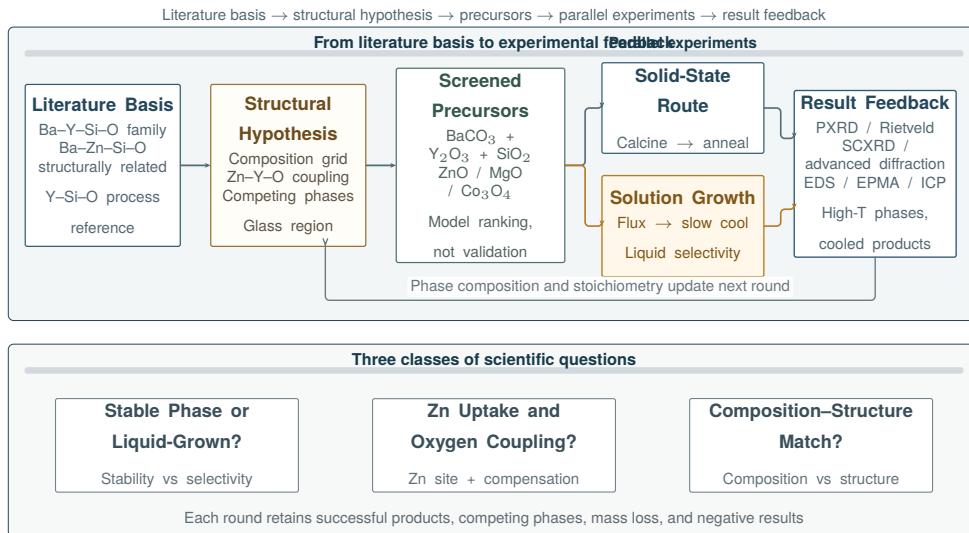


Figure 1: Research roadmap. The Ba-Y-Si-O, Ba-Zn-Si-O, and Y-Si-O systems are used first to establish structural relationships, competing phases, and process boundaries; phase-diagram and charge-balance analysis proposes candidate compositions, and the two-step inorganic retrosynthesis model supplies the precursor set; solid-state reaction and high-temperature solution growth probe the bulk phase region and the single-crystal structure, respectively, and the characterization results feed back into the next round of compositions and processes.

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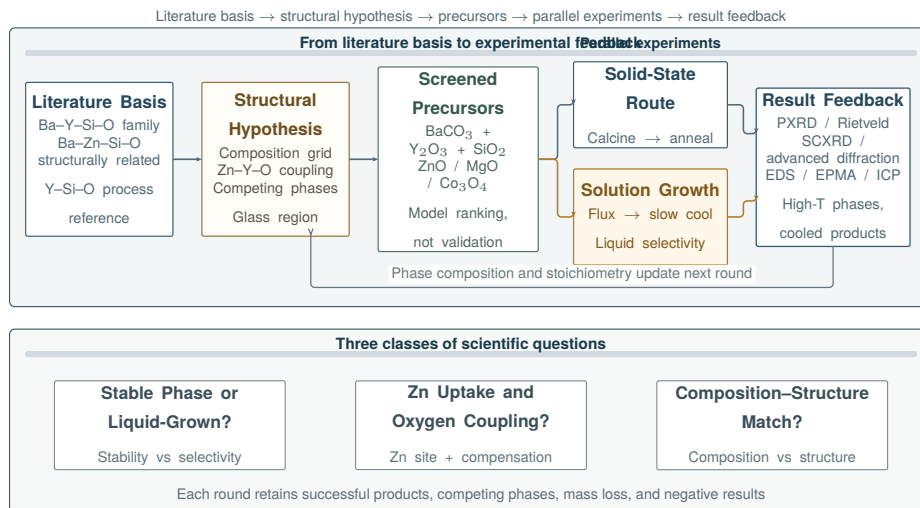


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Records were first merged by DOI; where a DOI was missing or several versions existed, they were cross-deduplicated by title, authors, year, and source, and author order, volume and pages, and publication status were checked. The search returned 63 candidate records, of which 51 independent

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An independent phase-region study combining EDS, cRED, synchrotron PXRD, and neutron refinement confirmed tetragonal $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ [18], giving the Zn-free Ba–Y silicates a structural reference that is independent of the tentative nomenclature. The single-crystal structure of

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Anonymous Review Working Group

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The non-centrosymmetric silicate $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ has been grown from a high-temperature solution and its single-crystal structure solved, but direct evidence for its starting composition, its bulk phase region, and any independent reproduction is still missing. Here we grade the available evidence and compare the published compositions, thermal histories, containers, cooling programs, and characterization results for the target phase, the Zn-free tetragonal Ba–Y silicate family, structurally related Ba–Zn–Si–O compounds, and Y–Si–O process-reference systems. The existing experiments show that phase formation near the target is governed simultaneously by the Ba/Y/Si ratio, the Zn–Y–oxygen stoichiometry, the flux proportion, and the cooling path, and that single-crystal structural evidence is no substitute for bulk phase-purity and measured-composition analysis. On this basis we propose four priorities: establishing a local phase diagram around $\text{Ba}_5\text{Y}_{12}\text{ZnSi}_8\text{O}_{40}$, testing the composition–structure relationship along $\text{Ba}_5\text{Y}_{13-x}\text{Zn}_x[\text{SiO}_4]_8\text{O}_{8.5-x/2}$, running solid-state reaction and high-temperature solution growth in parallel, and using Mg/Co analogs to isolate the effect of ionic-size differences. A local two-step inorganic retrosynthesis model recommends ZnO, MgO, and Co_3O_4 , respectively, together with BaCO_3 , Y_2O_3 , and SiO_2 , as the preferred starting-material sets for the Zn, Mg, and Co targets; these are precursor hypotheses awaiting verification, nothing more. The experiments most likely to yield discovery are not further trials at isolated temperatures but a closed feedback loop that ties together composition grids, phase quantification, single-crystal structure, measured stoichiometry, and process records.

1 Introduction

This review concerns the non-centrosymmetric barium yttrium zinc silicate $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$. We pursue three questions: whether the composition and structural assignment of the target phase can be confirmed; under which phase-formation conditions the phase can be obtained reproducibly; and whether a continuous composition–structure relationship connects the Zn-containing phase to the Zn-free tetragonal Ba–Y silicates. In the original report, crystals were grown from a high-temperature solution in an open system and the structure was solved by single-crystal X-ray diffraction (SCXRD) [1]. The structure combines a framework of highly coordinated rare-earth–oxygen polyhedra with isolated or composite silicate units, and it offers a reference point both for how complex silicates assemble and for the compositional design of non-centrosymmetric oxides.

The direct report of the target phase establishes only that the crystals exist and what their structure is; it gives no independently reproducible starting composition, no phase-formation range, and no route to phase-pure bulk samples [1]. We therefore do not stitch together a “best recipe” from the conditions used for structurally related compounds. Instead, we separate the questions that single-crystal growth, solid-state reaction, and phase-diagram screening can each answer, and we use that separation to design experiments that discriminate between alternative structural hypotheses. Literature values are quoted throughout with their stated limitations; structurally related systems and process-reference systems serve only to suggest experimental reference points.

1.1 Current Understanding of Related Systems

The tetragonal Ba–Y–Si–O silicate family. Early flux-growth experiments reported $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ as an accompanying crystalline phase, and a later series of 151 runs in the $\text{BaO–K}_2\text{O–Y}_2\text{O}_3\text{–SiO}_2\text{–MoO}_3$ system showed that the $\text{Y}_2\text{O}_3/\text{SiO}_2$ ratio, the MoO_3 content, and the cooling rate together govern which silicate structure type forms and how well the crystals grow [25, 42]. Recent single-crystal work has recast the related nominal composition as non-stoichiometric $\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$, and ceramic experiments showed that grinding in agate can introduce SiO_2 and push bulk samples off the nominal ratio [43, 44]. A separate phase-region study that combined energy-dispersive spectroscopy (EDS), continuous rotation electron diffraction (cRED), synchrotron powder X-ray diffraction (PXRD), and neutron diffraction confirmed $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ and

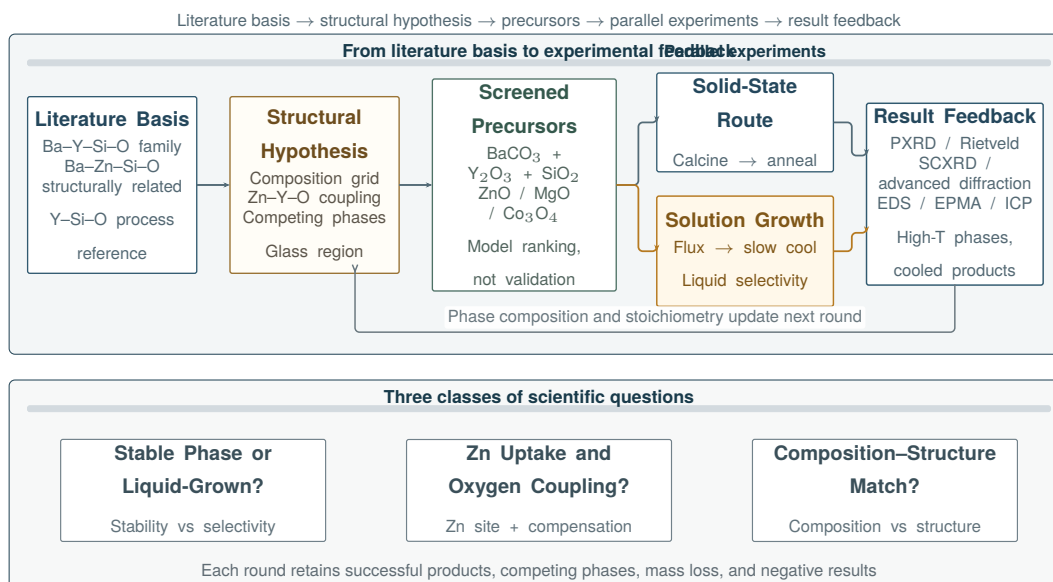


Figure 1: Research roadmap. The Ba–Y–Si–O, Ba–Zn–Si–O, and Y–Si–O systems are used first to establish structural relationships, competing phases, and process boundaries; phase-diagram and charge-balance analysis proposes candidate compositions, and the two-step inorganic retrosynthesis model supplies the precursor set; solid-state reaction and high-temperature solution growth probe the bulk phase region and the single-crystal structure, respectively, and the characterization results feed back into the next round of compositions and processes.

related only by element or function, with no crystallographic or synthetic link; phases predicted computationally but never realized experimentally; and aggregated abstracts or product information whose sources could not be traced. Glasses and glass-ceramics were treated as process-reference systems only where they clarify crystallization boundaries or competing phases; they support no inference about the structure of the target phase.

Records were first merged by DOI; where a DOI was missing or several versions existed, they were cross-duplicated by title, authors, year, and source, and author order, volume and pages, and publication status were checked. The search returned 63 candidate records, of which 51 independent publications survived deduplication and source verification. Because the target phase has only one direct primary paper, we claim neither comprehensive coverage nor a completed independent reproduction.

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We sorted the literature into five classes by how the subject of each study relates to the target phase: direct reports of the target compound; Zn-free members of the same tetragonal Ba–Y silicate family; structurally related compounds with a definite crystallographic relationship; synthesis studies usable as process references; and unrelated sources that cannot support any structural or recipe judgment. The classification applies to individual claims rather than to whole papers; the structural results and the process information from a single source may support conclusions of different scope.

For comparison, we extracted from each publication the starting materials and nominal ratios, pretreatment, mode of phase formation, temperature–time program, atmosphere and whether the system was open or closed, crucible or support, flux, cooling or growth program, and products and characterization methods. Numerical values stay attached to the original experimental record they came from and are never pooled across studies. An item the original text does not state explicitly is recorded as not reported, meaning that this check did not obtain the information, not that the original paper necessarily omitted it. Overall judgments favor results on which several studies agree for the same experimental item, and single-crystal structure, bulk phase composition, measured stoichiometry, and high-temperature stability are assessed separately.

Table 1: Compositional and structural evidence for the target compound and related Ba–Y silicates. Records are grouped by how the subject of each publication relates to the target phase; neither the Zn-free studies nor the structurally related studies constitute an independent reproduction of the Zn-containing target phase.

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$\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$	[1]	Single-crystal X-ray structure determination [1]	Direct report of the target compound	Established: Zn-containing target phase and its route type; open: independent reproduction, bulk phase-pure window, correspondence to the Zn-free family
$\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ (tentative; x kept variable here)	[25, 42]	Reported as an accompanying phase; systematic phase-formation screening [25, 42]	Same-family tetragonal Ba–Y silicate	Established: historical name and composition-screening results; open: exact stoichiometry, structural model, relation to the target phase
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ (non-stoichiometric formula kept here)	[43]	Single-crystal revision to a non-stoichiometric modulated structure [43]	Same-family tetragonal Ba–Y silicate	Established: current structural reading of the historical formula; open: whether Zn can enter this family
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ ceramics	[44]	Bulk phase composition and microstructure [44]	Same-family tetragonal Ba–Y silicate	Established: bulk characterization requirements for the non-stoichiometric family; open: separating secondary phases and extraneous SiO_2 from the intrinsic phase region
$\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$	[18]	EDS, cRED, synchrotron PXRD, and neutron refinement [18]	Same-family tetragonal Ba–Y silicate	Established: independently confirmed tetragonal Ba–Y structure and phase-region reference; open: site correspondence to the Zn-containing target phase
$\text{BaY}_{16}\text{Si}_4\text{O}_{33}$	[30]	Single-crystal structure with isolated orthosilicate units [30]	Structurally related compound	Established: structural reference for Ba–Y orthosilicates; open: no evidence that it is the same phase as the Zn-containing target

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Direct reports of the target compound. The formula must match $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$, and the primary study must give both the preparation and the structure determination. At present only the open-system high-temperature solution growth and single-crystal X-ray study of Ababaikeri et al. qualifies [1]. It provides the sole direct experimental starting point, but it has not established an independently reproducible bulk phase-pure window.

Same-family tetragonal Ba–Y silicates. These studies contain no Zn but bear directly on the historical nomenclature, the structural revision, or the phase-region relationships of the tetragonal Ba–Y silicates. Early work reported the tentative accompanying phase $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ [25]; modern single-crystal work revised the related composition to a non-stoichiometric modulated structure [43]; and an independent study confirmed tetragonal $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ by several diffraction and compositional methods [18]. These results constrain the historical formulas, the composition range, and the competing phases, but they are no substitute for reproducing the Zn-containing target phase.

Synthesis Conditions of $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ and Its Structural Relatives: Evidence Grading, Route Matrix, and Transferability Limits

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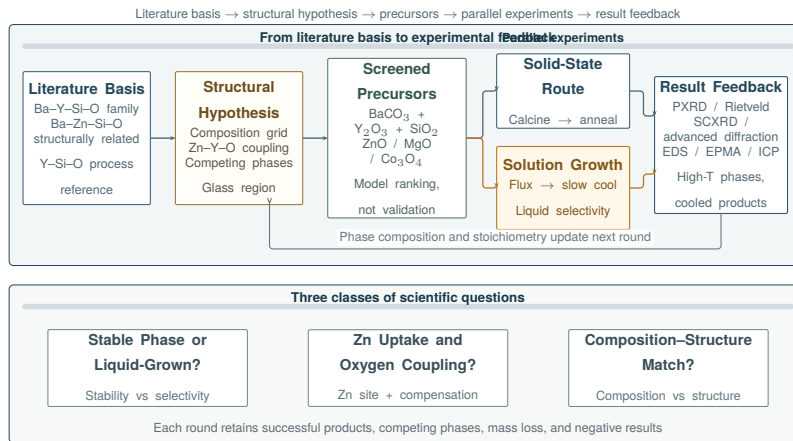


Figure 1. Research roadmap. The Ba–Y–Si–O, Ba–Zn–Si–O, and Y–Si–O systems are used first to establish structural relationships, competing phases, and process boundaries; phase-diagram and charge-balance analysis proposes candidate compositions, and the two-step inorganic retrosynthesis model supplies the precursor set; solid-state reaction and high-temperature solution growth probe the bulk phase region and the single-crystal structure, respectively, and the characterization results feed back into the next round of compositions and processes.

experimental conditions, and finally propose four research directions: a local phase diagram, Zn–Y–oxygen-stoichiometry coupling, solid-state reaction run in parallel with high-temperature solution growth, and Mg/Co analogs. The two-step inorganic retrosynthesis model used here generates only a set of precursors to be tested; temperatures, atmospheres, crucibles, and characterization methods are still fixed from the literature and from materials-chemistry constraints.

Figure 1 summarizes the line of argument. Related systems are used first to establish structural relationships, competing phases, and process boundaries; phase-diagram and charge-balance analysis then proposes candidate compositions, from which the retrosynthesis model derives the precursor set; solid-state reaction and high-temperature solution growth test the bulk phase region and the single-crystal structure, respectively; and composition and process are adjusted round by round on the basis of the phase composition, the measured stoichiometry, and the structure refinement. Each round of experiments should eliminate at least one structural or phase-region hypothesis rather than chase an unverified “best recipe”.

2 Literature Search and Evidence Assessment

2.1 Search Scope

The search began from the target formula in both its structural and its elemental spelling, using Ba₅Y₁₂Zn[O(SiO₄)₈], Ba₅Y₁₂ZnSi₈O₄₀, BYZSO, and barium yttrium zinc silicate, combined with the keywords space group, crystal structure, flux, solid-state, phase diagram, and crystal growth. We then followed the literature in three directions:

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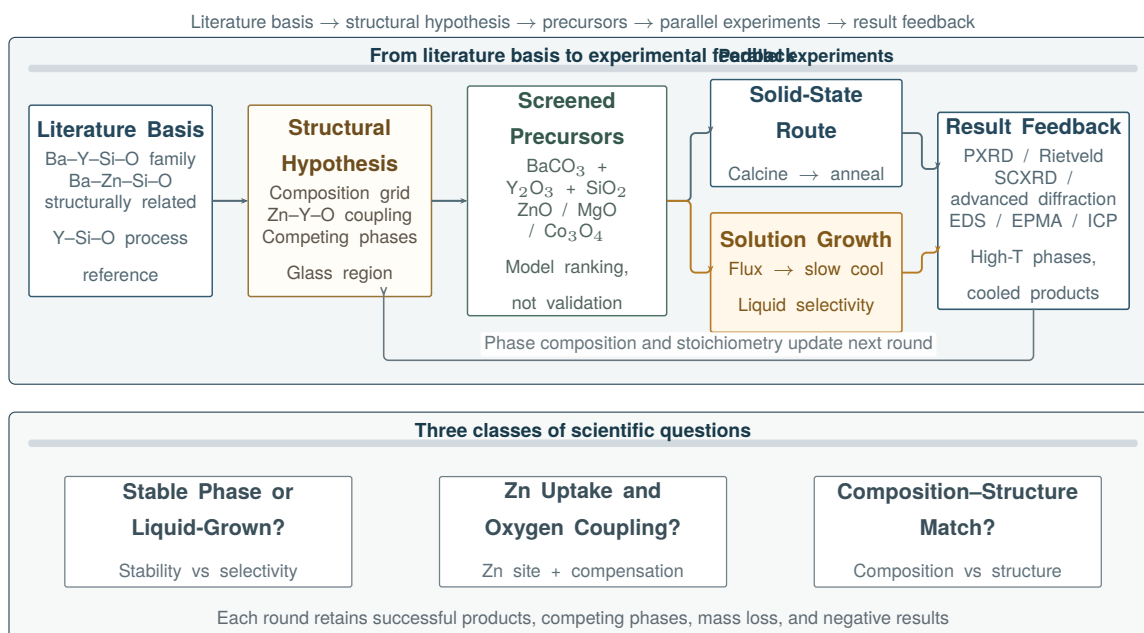


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Ba₅Y₁₂Zn[O(SiO₄)]₈ 及结构相关化合物的 合成条件：证据分级、路线矩阵与迁移边界

匿名综述工作组

摘要

非中心对称硅酸盐 Ba₅Y₁₂Zn[O(SiO₄)]₈ 已由高温溶液法获得并完成单晶结构解析，但其投料组成、块体相区和独立复现仍缺少直接证据。本文围绕目标相、无 Zn 的四方 Ba–Y 硅酸盐系列、Ba–Zn–Si–O 结构相关化合物及 Y–Si–O 工艺参照体系，对已发表的组成、热历史、容器、冷却和表征结果进行分级比较。现有实验显示，目标附近的相形成行为同时受 Ba/Y/Si 比、Zn–Y–氧计量、助熔剂比例和冷却路径影响；单晶结构证据不能替代块体相纯度和实际组成分析。基于这些结果，本文提出四项优先工作：建立 Ba₅Y₁₂ZnSi₈O₄₀ 附近的局部相图，检验 Ba₅Y_{13–x}Zn_x[SiO₄]₈O_{8.5–x/2} 的组成–结构关系，并行比较固相反应与高温溶液法晶体生长，以及用 Mg/Co 类比区分尺寸差异。本地两步无机逆合成模型为 Zn、Mg、Co 目标分别推荐以 ZnO、MgO、Co₃O₄ 配合 BaCO₃、Y₂O₃ 和 SiO₂ 的首选原料集合；这些结果仅作为待验证的前驱体假设。最有利于科学发现的实验不是继续尝试孤立温度，而是将组成网格、相定量、单晶结构、实际化学计量和工艺记录组成可反馈的实验闭环。

1 引言

本文的研究对象为非中心对称钡–钇–锌硅酸盐 Ba₅Y₁₂Zn[O(SiO₄)]₈。研究目标包括三方面：确认目标相的组成与结构归属；厘清可重复获得该相的相形成条件；判断含 Zn 相与无 Zn 四方 Ba–Y 硅酸盐之间是否存在连续的组成–结构关系。原始报道在开放体系中以高温溶液法生晶体生长体，并由单晶 X 射线衍射 (SCXRD) 确定结构 [1]。该结构兼具高配位稀土–氧骨架和孤立或复合硅酸根单元，可为复杂硅酸盐的结构组装和非中心对称氧化物的组成设计提供参照。

目标相的直接报道仅明确了晶体的存在及其结构，尚未给出可独立复现的投料组成、相形成范围或块体相纯样品的制备方法 [1]。因此，本文不由结构相关化合物的条件组合出一条“最佳配方”，而是区分单晶生长、固相反应和相图筛查各自能够解决的问题，并据此设计可区

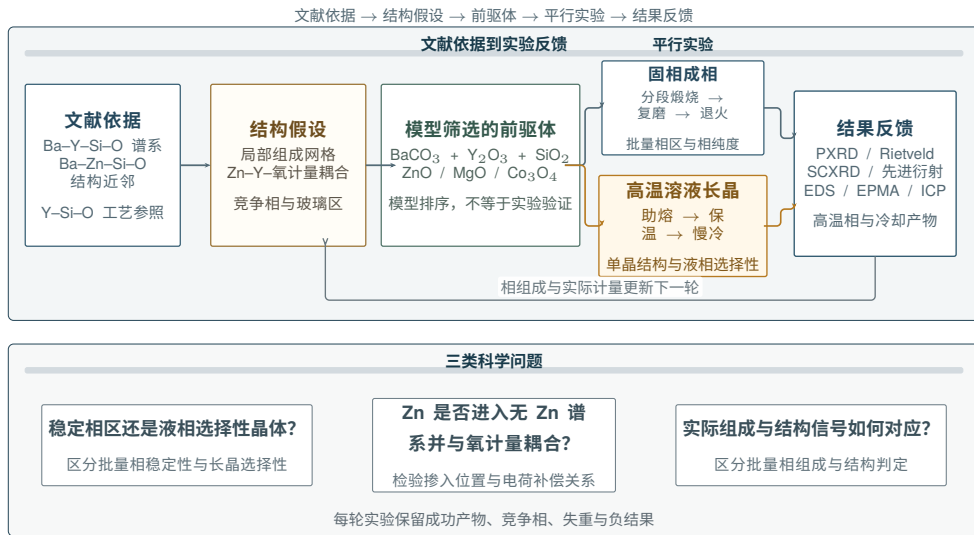


图 1: 本文的研究路线。 Ba-Y-Si-O、Ba-Zn-Si-O 和 Y-Si-O 体系先用于确定结构关系、竞争相和工艺边界；相图与电荷平衡分析提出候选组成，两步无机逆合成模型给出前驱体集合；固相反应与高温溶液法晶体生长分别考察块体相区和单晶结构，表征结果用于调整后续的组成和工艺。

2 文献检索与证据评价

2.1 检索范围

检索以目标化学式和元素式为起点，采用 $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ 、 $\text{Ba}_5\text{Y}_{12}\text{ZnSi}_8\text{O}_{40}$ 、BYZSO 和 barium yttrium zinc silicate 等写法，并组合空间群、晶体结构、flux、solid-state、phase diagram 和 crystal growth 等关键词。随后沿三个方向追溯相关文献：(i) 无 Zn 四方 Ba-Y 硅酸盐的命名与结构系列；(ii) 具有明确结构证据的 Ba-Zn-Si-O 化合物；(iii) 报道高温溶液法、固相反应、机械活化或提拉法工艺的 Y-Si-O 体系。检索先查本地材料全文库，再通过 OpenAlex、Crossref、出版社页面、DOI 解析页面、IUCr、COD、RSC 正文与补充材料 (SI) 以及机构知识库核对元数据和实验段落。检索截止日期为 2026-09-01。数据库或接口未命中仅说明该渠道未返回结果，不能据此断言文献不存在。

2.2 纳入、排除与去重

满足下列任一条件的文献予以纳入：直接报道目标相的制备或结构；重新审定 Ba-Y 四方硅酸盐系列的结构、相区或组成系列；为 Ba-Zn-Si-O 或 Y-Si-O 结构相关体系提供可追溯的结构数据或合成条件。综述类文献仅用于统一术语和追溯原始论文，具体实验条件均以原始研究为准。下列资料予以排除：仅元素或功能相近而无晶体学或合成联系的材料；仅有计算预测而未经实验实现的相；来源无法追溯的聚合摘要或产品信息。玻璃和微晶玻璃仅在说明晶化边界或竞争相时作为工艺参照体系，不能据此推断目标相的结构。

表 1: 目标化合物及相关 Ba–Y 硅酸盐的组成与结构依据。表中记录按文献对象与目标相的关系分类；无 Zn 研究和结构相关研究均不构成含 Zn 目标相的独立复现。

化学式（原文 → 本文写法）	文献	主要结构证据	与目标相 的关系	结论与待研究问题
$\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$	[1]	单晶 X 射线结构鉴定 [1]	目标化合物直接文献	结论：含 Zn 目标相及其合成路线类型；待研究：独立复现、块体相纯窗口及与无 Zn 系列的对应
$\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ （暂定）（本文：保留变量 x ）	[25, 42]	伴生相报道与系统相形成筛查 [25, 42]	同类 Ba–Y 四方硅酸盐	结论：历史命名及组成筛查结果；待研究：精确化学计量、结构模型及与目标相的关系
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ （本文：保留非化学计量式）	[43]	单晶非化学计量调制结构重新审定 [43]	同类 Ba–Y 四方硅酸盐	结论：历史式的现代结构解释；待研究：Zn 是否可进入该系列
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ 陶瓷	[44]	块体相组成与显微结构分析 [44]	同类 Ba–Y 四方硅酸盐	结论：非化学计量系列的块体表征需求；待研究：第二相、外来 SiO_2 与本征相区的区分
$\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$	[18]	EDS、cRED、同步辐射 PXRD 与中子精修 [18]	同类 Ba–Y 四方硅酸盐	结论：独立确认的四方 Ba–Y 结构与相区参照；待研究：与含 Zn 目标相的位点对应
$\text{BaY}_{16}\text{Si}_4\text{O}_{33}$	[30]	单晶结构及孤立正硅酸根单元 [30]	结构相关化合物	结论：Ba–Y 正硅酸盐的结构参照；待研究：缺乏与含 Zn 目标相同相的依据

线类型和单晶结构归属，但不能证明跨实验室复现或块体相纯窗口，也未完全厘清氧计量表达式、Zn 位点及其与无 Zn 四方硅酸盐系列的关系。

表 1 分列原文写法与本文的排版形式。后者仅统一符号、下标和括号，不将变组成式改写为定组成式，也不依据元素守恒自行修改已发表的化学式。

3.3 四方硅酸盐系列的重新审定

历史名称的沿用不等于结构的延续。早期助熔剂法生长研究将 $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ 记为暂定伴生相 [25]；后续的系统助熔剂法研究将其纳入多变量相形成筛查，但对象仍为无 Zn 的 Ba–Y 四方硅酸盐 [42]。现代单晶研究将相关名义组成重新审定为 $\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ 非化学计量调制结构 [43]；相应的陶瓷研究表明，块体样品还须检查第二相和研磨介质引入的 SiO_2 [44]。因此，历史名称须依据结构模型和块体相组成重新核对；上述研究均未直接观察到精确的含 Zn 目标化学式。

独立相区研究结合 EDS、cRED、同步辐射 PXRD 和中子精修，确认了四方 $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ [18]，为无 Zn 的 Ba–Y 硅酸盐提供了有别于暂定命名的结构参照。 $\text{BaY}_{16}\text{Si}_4\text{O}_{33}$ 的单晶结构则给出了含孤立正硅酸根单元的 Ba–Y 参照结构 [30]，可用于检查多面体与硅酸根的连接方

Ba₅Y₁₂Zn[O(SiO₄)]₈ 及结构相关化合物的 合成条件：证据分级、路线矩阵与迁移边界

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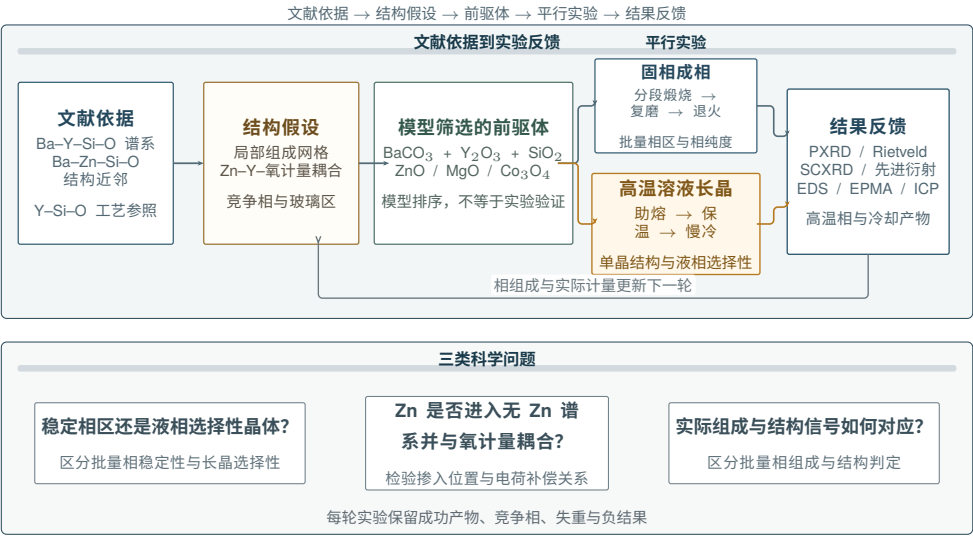


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满足下列任一条件的文献予以纳入: 直接报道目标相的制备或结构; 重新审定 Ba-Y 四方硅酸盐系列的结构、相区或组成系列; 为 Ba-Zn-Si-O 或 Y-Si-O 结构相关体系提供可追溯的结构数据或合成条件。综述类文献仅用于统一术语和追溯原始论文, 具体实验条件均以原始研究为准。下列资料予以排除: 仅元素或功能相近而无晶体学或合成联系的材料; 仅有计算预测而未经实验实现的相; 来源无法追溯的聚合摘要或产品信息。玻璃和微晶玻璃仅在说明晶化边界或竞争相时作为工艺参照体系, 不能据此推断目标相的结构。

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化学式（原文 → 本文写法）	文献	主要结构证据	与目标相 的关系	结论与待研究问题
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$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ 陶瓷	[44]	块体相组成与显微结构分析 [44]	同类 Ba–Y 四方硅酸盐	结论：非化学计量系列的块体表征需求；待研究：第二相、外来 SiO_2 与本征相区的区分
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匿名综述工作组

摘要

非中心对称硅酸盐 Ba₅Y₁₂Zn[O(SiO₄)]₈ 已由高温溶液法获得并完成单晶结构解析，但其投料组成、块体相区和独立复现仍缺少直接证据。本文围绕目标相、无 Zn 的四方 Ba–Y 硅酸盐系列、Ba–Zn–Si–O 结构相关化合物及 Y–Si–O 工艺参照体系，对已发表的组成、热历史、容器、冷却和表征结果进行分级比较。现有实验显示，目标附近的相形成行为同时受 Ba/Y/Si 比、Zn–Y–氧计量、助熔剂比例和冷却路径影响；单晶结构证据不能替代块体相纯度和实际组成分析。基于这些结果，本文提出四项优先工作：建立 Ba₅Y₁₂ZnSi₈O₄₀ 附近的局部相图，检验 Ba₅Y_{13–x}Zn_x[SiO₄]₈O_{8.5–x/2} 的组成–结构关系，并行比较固相反应与高温溶液法晶体生长，以及用 Mg/Co 类比区分尺寸差异。本地两步无机逆合成模型为 Zn、Mg、Co 目标分别推荐以 ZnO、MgO、Co₃O₄ 配合 BaCO₃、Y₂O₃ 和 SiO₂ 的首选原料集合；这些结果仅作为待验证的前驱体假设。最有利于科学发现的实验不是继续尝试孤立温度，而是将组成网格、相定量、单晶结构、实际化学计量和工艺记录组成可反馈的实验闭环。

1 引言

本文的研究对象为非中心对称钡–钇–锌硅酸盐 Ba₅Y₁₂Zn[O(SiO₄)]₈。研究目标包括三方面：确认目标相的组成与结构归属；厘清可重复获得该相的相形成条件；判断含 Zn 相与无 Zn 四方 Ba–Y 硅酸盐之间是否存在连续的组成–结构关系。原始报道在开放体系中以高温溶液法生晶体生长体，并由单晶 X 射线衍射 (SCXRD) 确定结构 [1]。该结构兼具高配位稀土–氧骨架和孤立或复合硅酸根单元，可为复杂硅酸盐的结构组装和非中心对称氧化物的组成设计提供参照。

目标相的直接报道仅明确了晶体的存在及其结构，尚未给出可独立复现的投料组成、相形成范围或块体相纯样品的制备方法 [1]。因此，本文不由结构相关化合物的条件组合出一条“最佳配方”，而是区分单晶生长、固相反应和相图筛查各自能够解决的问题，并据此设计可区

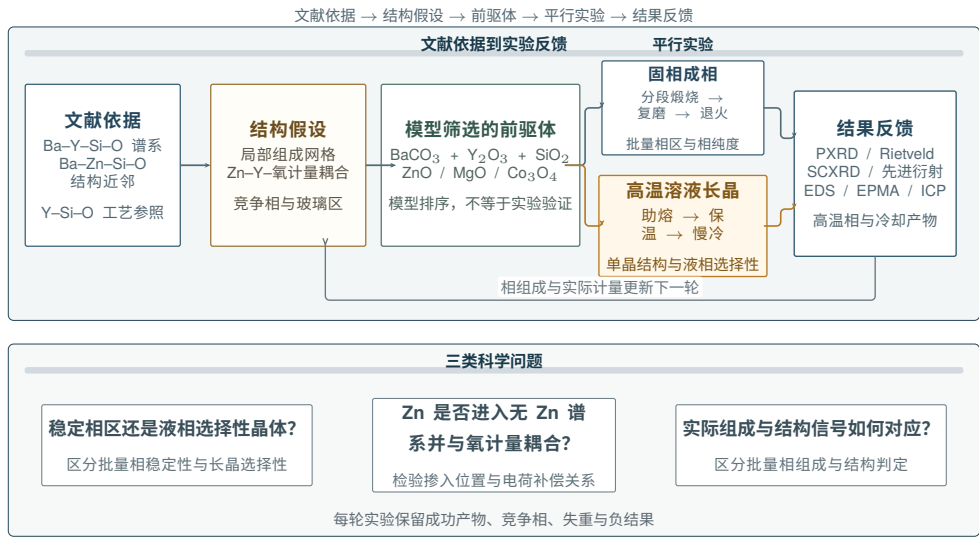


图 1. 本文的研究路线。 Ba-Y-Si-O、Ba-Zn-Si-O 和 Y-Si-O 体系先用于确定结构关系、竞争相和工艺边界；相图与电荷平衡分析提出候选组成，两步无机逆合成模型给出前驱体集合；固相反应与高温溶液法晶体生长分别考察块体相区和单晶结构，表征结果用于调整后续的组成和工艺。

2 文献检索与证据评价

2.1 检索范围

检索以目标化学式和元素式为起点，采用 $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ 、 $\text{Ba}_5\text{Y}_{12}\text{ZnSi}_8\text{O}_{40}$ 、BYZSO 和 barium yttrium zinc silicate 等写法，并组合空间群、晶体结构、flux、solid-state、phase diagram 和 crystal growth 等关键词。随后沿三个方向追溯相关文献：(i) 无 Zn 四方 Ba-Y 硅酸盐的命名与结构系列；(ii) 具有明确结构证据的 Ba-Zn-Si-O 化合物；(iii) 报道高温溶液法、固相反应、机械活化或提拉法工艺的 Y-Si-O 体系。检索先查本地材料全文库，再通过 OpenAlex、Crossref、出版社页面、DOI 解析页面、IUCr、COD、RSC 正文与补充材料 (SI) 以及机构知识库核对元数据和实验段落。检索截止日期为 2026-09-01。数据库或接口未命中仅说明该渠道未返回结果，不能据此断言文献不存在。

2.2 纳入、排除与去重

满足下列任一条件的文献予以纳入：直接报道目标相的制备或结构；重新审定 Ba-Y 四方硅酸盐系列的结构、相区或组成系列；为 Ba-Zn-Si-O 或 Y-Si-O 结构相关体系提供可追溯的结构数据或合成条件。综述类文献仅用于统一术语和追溯原始论文，具体实验条件均以原始研究为准。下列资料予以排除：仅元素或功能相近而无晶体学或合成联系的材料；仅有计算预测而未经实验实现的相；来源无法追溯的聚合摘要或产品信息。玻璃和微晶玻璃仅在说明晶化边界或竞争相时作为工艺参照体系，不能据此推断目标相的结构。

表 1. 目标化合物及相关 Ba–Y 硅酸盐的组成与结构依据。表中记录按文献对象与目标相的关系分类；无 Zn 研究和结构相关研究均不构成含 Zn 目标相的独立复现。

化学式（原文 → 本文写法）	文献	主要结构证据	与目标相 的关系	结论与待研究问题
$\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$	[1]	单晶 X 射线结构鉴定 [1]	目标化合物直接文献	结论：含 Zn 目标相及其合成路线类型；待研究：独立复现、块体相纯窗口及与无 Zn 系列的对应
$\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ (暂定)（本文：保留变量 x ）	[25, 42]	伴生相报道与系统相形成筛查 [25, 42]	同类 Ba–Y 四方硅酸盐	结论：历史命名及组成筛查结果；待研究：精确化学计量、结构模型及与目标相的关系
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ （本文：保留非化学计量式）	[43]	单晶非化学计量调制结构重新审定 [43]	同类 Ba–Y 四方硅酸盐	结论：历史式的现代结构解释；待研究：Zn 是否可进入该系列
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ 陶瓷	[44]	块体相组成与显微结构分析 [44]	同类 Ba–Y 四方硅酸盐	结论：非化学计量系列的块体表征需求；待研究：第二相、外来 SiO_2 与本征相区的区分
$\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$	[18]	EDS、cRED、同步辐射 PXRD 与中子精修 [18]	同类 Ba–Y 四方硅酸盐	结论：独立确认的四方 Ba–Y 结构与相区参照；待研究：与含 Zn 目标相的位点对应
$\text{BaY}_{16}\text{Si}_4\text{O}_{33}$	[30]	单晶结构及孤立正硅酸根单元 [30]	结构相关化合物	结论：Ba–Y 正硅酸盐的结构参照；待研究：缺乏与含 Zn 目标相同相的依据

线类型和单晶结构归属，但不能证明跨实验室复现或块体相纯窗口，也未完全厘清氧计量表达式、Zn 位点及其与无 Zn 四方硅酸盐系列的关系。

表 1 分列原文写法与本文的排版形式。后者仅统一符号、下标和括号，不将变组成式改写为定组成式，也不依据元素守恒自行修改已发表的化学式。

3.3 四方硅酸盐系列的重新审定

历史名称的沿用不等于结构的延续。早期助熔剂法生长研究将 $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ 记为暂定伴生相 [25]；后续的系统助熔剂法研究将其纳入多变量相形成筛查，但对象仍为无 Zn 的 Ba–Y 四方硅酸盐 [42]。现代单晶研究将相关名义组成重新审定为 $\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ 非化学计量调制结构 [43]；相应的陶瓷研究表明，块体样品还须检查第二相和研磨介质引入的 SiO_2 [44]。因此，历史名称须依据结构模型和块体相组成重新核对；上述研究均未直接观察到精确的含 Zn 目标化学式。

独立相区研究结合 EDS、cRED、同步辐射 PXRD 和中子精修，确认了四方 $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ [18]，为无 Zn 的 Ba–Y 硅酸盐提供了有别于暂定命名的结构参照。 $\text{BaY}_{16}\text{Si}_4\text{O}_{33}$ 的单晶结构则给出了含孤立正硅酸根单元的 Ba–Y 参照结构 [30]，可用于检查多面体与硅酸根的连接方

Ba₅Y₁₂Zn[O(SiO₄)]₈ 及结构相关化合物的 合成条件：证据分级、路线矩阵与迁移边界

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摘要

非中心对称硅酸盐 Ba₅Y₁₂Zn[O(SiO₄)]₈ 已由高温溶液法获得并完成单晶结构解析，但其投料组成、块体相区和独立复现仍缺少直接证据。本文围绕目标相、无 Zn 的四方 Ba–Y 硅酸盐系列、Ba–Zn–Si–O 结构相关化合物及 Y–Si–O 工艺参照体系，对已发表的组成、热历史、容器、冷却和表征结果进行分级比较。现有实验显示，目标附近的相形成行为同时受 Ba/Y/Si 比、Zn–Y–氧计量、助熔剂比例和冷却路径影响；单晶结构证据不能替代块体相纯度和实际组成分析。基于这些结果，本文提出四项优先工作：建立 Ba₅Y₁₂ZnSi₈O₄₀ 附近的局部相图，检验 Ba₅Y_{13–x}Zn_x[SiO₄]₈O_{8.5–x/2} 的组成–结构关系，并行比较固相反应与高温溶液法晶体生长，以及用 Mg/Co 类比区分尺寸差异。本地两步无机逆合成模型为 Zn、Mg、Co 目标分别推荐以 ZnO、MgO、Co₃O₄ 配合 BaCO₃、Y₂O₃ 和 SiO₂ 的首选原料集合；这些结果仅作为待验证的前驱体假设。最有利于科学发现的实验不是继续尝试孤立温度，而是将组成网格、相定量、单晶结构、实际化学计量和工艺记录组成可反馈的实验闭环。

1 引言

本文的研究对象为非中心对称钇–钇–锌硅酸盐 Ba₅Y₁₂Zn[O(SiO₄)]₈。研究目标包括三方面：确认目标相的组成与结构归属；厘清可重复获得该相的相形成条件；判断含 Zn 相与无 Zn 四方 Ba–Y 硅酸盐之间是否存在连续的组成–结构关系。原始报道在开放体系中以高温溶液法生晶体生长体，并由单晶 X 射线衍射 (SCXRD) 确定结构 [1]。该结构兼具高配位稀土–氧骨架和孤立或复合硅酸根单元，可为复杂硅酸盐的结构组装和非中心对称氧化物的组成设计提供参照。

目标相的直接报道仅明确了晶体的存在及其结构，尚未给出可独立复现的投料组成、相形成范围或块体相纯样品的制备方法 [1]。因此，本文不由结构相关化合物的条件组合出一条“最佳配方”，而是区分单晶生长、固相反应和相图筛查各自能够解决的问题，并据此设计可区分不同结构假设的实验。全文保留文献中的数值及其限制条件；结构相关体系与工艺参照体系仅用于提出实验参照。

1.1 相关体系的研究现状

Ba–Y–Si–O 四方硅酸盐系列。 早期助熔剂法生长实验将 Ba_{5+x}Y₁₃Si₈O₄₁ 报道为伴生晶相；随后的 151 组 BaO–K₂O–Y₂O₃–SiO₂–MoO₃ 体系实验表明，Y₂O₃/SiO₂ 比、MoO₃ 用量和冷却速率共同影响硅酸盐的结构类型与晶体质量 [25, 42]。现代单晶研究将相关名义组成重新审定为非化学计量

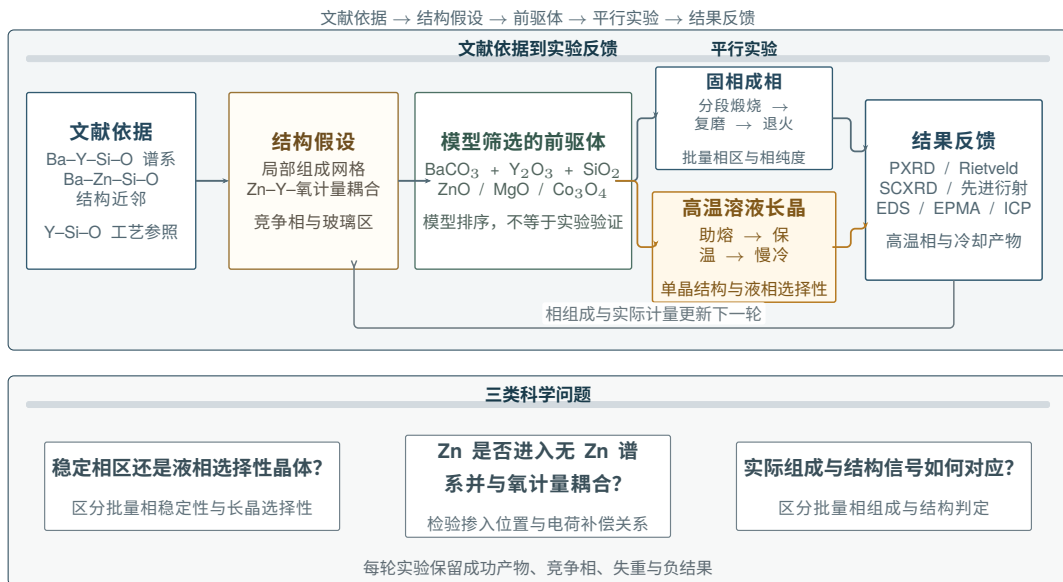


图 1: 本文的研究路线。 Ba-Y-Si-O、Ba-Zn-Si-O 和 Y-Si-O 体系先用于确定结构关系、竞争相和工艺边界；相图与电荷平衡分析提出候选组成，两步无机逆合成模型给出前驱体集合；固相反应与高温溶液法晶体生长分别考察块体相区和单晶结构，表征结果用于调整后续的组成和工艺。

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满足下列任一条件的文献予以纳入：直接报道目标相的制备或结构；重新审定 Ba-Y 四方硅酸盐系列的结构、相区或组成系列；为 Ba-Zn-Si-O 或 Y-Si-O 结构相关体系提供可追溯的结构数据或合成条件。综述类文献仅用于统一术语和追溯原始论文，具体实验条件均以原始研究为准。下列资料予以排除：仅元素或功能相近而无晶体学或合成联系的材料；仅有计算预测而未经实验实现的相；来源无法追溯的聚合摘要或产品信息。玻璃和微晶玻璃仅在说明晶化边界或竞争相时作为工艺参照体系，不能据此推断目标相的结构。

记录先按 DOI 合并；无 DOI 或存在多个版本时，依据题名、作者、年份和来源交叉去重，并核对作者顺序、卷页和出版状态。检索共得到 63 条候选记录，去重和来源核验后保留 51 篇独立文献。目标相仅有一篇直接原始论文，因此本文不宣称覆盖全面，也不宣称已完成独立复现。

2.3 文献相关程度与条件提取

本文按研究对象与目标相的关系将文献分为五类：目标化合物的直接报道、无 Zn 的同类 Ba-Y 四方硅酸盐、具有明确晶体学关系的结构相关化合物、可作工艺参照的合成研究，以及不能支持结构或配方判断的非相关资料。分类针对具体论断而非整篇论文，同一来源的结构结果和工艺信息可支持不同范围的解释。

为便于比较，逐篇提取原料及名义配比、前处理、相形成方式、温度-时间、气氛与体系开放性、坩埚或承载体、助熔剂、冷却或生长程序以及产物与表征方法。数值仅保留在对应的原始实验记录中，不跨研究合并。原文未明确给出的项目记为未报道，表示本次核查未取得该信息，并不表示原文

表 1: 目标化合物及相关 Ba–Y 硅酸盐的组成与结构依据。表中记录按文献对象与目标相的关系分类；无 Zn 研究和结构相关研究均不构成含 Zn 目标相的独立复现。

化学式（原文 → 本文写法）	文献	主要结构证据	与目标相的关系	结论与待研究问题
$\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$	[1]	单晶 X 射线结构鉴定 [1]	目标化合物直接文献	结论：含 Zn 目标相及其合成路线类型；待研究：独立复现、块体相纯度窗口及与无 Zn 系列的对应
$\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ （暂定）（本文：保留变量 x ）	[25, 42]	伴生相报道与系统相形成筛查 [25, 42]	同类 Ba–Y 四方硅酸盐	结论：历史命名及组成筛查结果；待研究：精确化学计量、结构模型及与目标相的关系
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ （本文：保留非化学计量式）	[43]	单晶非化学计量调制结构重新审定 [43]	同类 Ba–Y 四方硅酸盐	结论：历史式的现代结构解释；待研究：Zn 是否可进入该系列
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ 陶瓷	[44]	块体相组成与显微结构分析 [44]	同类 Ba–Y 四方硅酸盐	结论：非化学计量系列的块体表征需求；待研究：第二相、外来 SiO_2 与本征相区的区分
$\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$	[18]	EDS、cRED、同步辐射 PXRD 与中子精修 [18]	同类 Ba–Y 四方硅酸盐	结论：独立确认的四方 Ba–Y 结构与相区参照；待研究：与含 Zn 目标相的位点对应
$\text{BaY}_{16}\text{Si}_4\text{O}_{33}$	[30]	单晶结构及孤立正硅酸根单元 [30]	结构相关化合物	结论：Ba–Y 正硅酸盐的结构参照；待研究：缺乏与含 Zn 目标相同相的依据

4 文献相关程度与可比范围

4.1 分类依据

本文按研究对象与目标相的关系组织文献，不将整篇论文简单归入单一类别。同一研究可分别为结构、合成路线或表征方法提供依据，每条结论仅适用于该文献直接观察的对象。文献分为五类：目标化合物的直接报道、无 Zn 的同类 Ba–Y 四方硅酸盐、具有明确晶体学关系的结构相关化合物、可作工艺参照的合成研究，以及不能支持目标结构或配方判断的非相关资料。分类用于说明结论的适用范围，不评价论文质量。

目标化合物的直接报道。 化学式须与 $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ 一致，且原始研究须同时给出制备过程和结构鉴定。目前仅 Ababaikeri 等报道的开放体系高温溶液法生长与单晶 X 射线结构研究符合这一条件 [1]。该研究提供了唯一的直接实验起点，但尚未建立可独立复现的块体相纯度窗口。

同类 Ba–Y 四方硅酸盐。 此类研究不含 Zn，但直接涉及四方 Ba–Y 硅酸盐的历史命名、结构重新审定或相区关系。早期工作报道了暂定的 $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ 伴生相 [25]；现代单晶研究将相关组成重新审定为非化学计量调制结构 [43]；独立研究则以多种衍射和成分分析方法确认了四方 $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ [18]。这些结果可限定历史化学式、组成范围和竞争相，但不能替代含 Zn 目标相的复现。

Ba₅Y₁₂Zn[O(SiO₄)]₈ 及结构相关化合物的 合成条件：证据分级、路线矩阵与迁移边界

匿名综述工作组

摘要

非中心对称硅酸盐 Ba₅Y₁₂Zn[O(SiO₄)]₈ 已由高温溶液法获得并完成单晶结构解析，但其投料组成、块体相区和独立复现仍缺少直接证据。本文围绕目标相、无 Zn 的四方 Ba–Y 硅酸盐系列、Ba–Zn–Si–O 结构相关化合物及 Y–Si–O 工艺参照体系，对已发表的组成、热历史、容器、冷却和表征结果进行分级比较。现有实验显示，目标附近的相形成行为同时受 Ba/Y/Si 比、Zn–Y–氧计量、助熔剂比例和冷却路径影响；单晶结构证据不能替代块体相纯度和实际组成分析。基于这些结果，本文提出四项优先工作：建立 Ba₅Y₁₂ZnSi₈O₄₀ 附近的局部相图，检验 Ba₅Y_{13–x}Zn_x[SiO₄]₈O_{8.5–x/2} 的组成–结构关系，并行比较固相反应与高温溶液法晶体生长，以及用 Mg/Co 类比区分尺寸差异。本地两步无机逆合成模型为 Zn、Mg、Co 目标分别推荐以 ZnO、MgO、Co₃O₄ 配合 BaCO₃、Y₂O₃ 和 SiO₂ 的首选原料集合；这些结果仅作为待验证的前驱体假设。最有利于科学发现的实验不是继续尝试孤立温度，而是将组成网格、相定量、单晶结构、实际化学计量和工艺记录组成可反馈的实验闭环。

1 引言

本文的研究对象为非中心对称钡–钇–锌硅酸盐 Ba₅Y₁₂Zn[O(SiO₄)]₈。研究目标包括三方面：确认目标相的组成与结构归属；厘清可重复获得该相的相形成条件；判断含 Zn 相与无 Zn 四方 Ba–Y 硅酸盐之间是否存在连续的组成–结构关系。原始报道在开放体系中以高温溶液法生晶体生长体，并由单晶 X 射线衍射 (SCXRD) 确定结构 [1]。该结构兼具高配位稀土–氧骨架和孤立或复合硅酸根单元，可为复杂硅酸盐的结构组装和非中心对称氧化物的组成设计提供参照。

目标相的直接报道仅明确了晶体的存在及其结构，尚未给出可独立复现的投料组成、相形成范围或块体相纯样品的制备方法 [1]。因此，本文不由结构相关化合物的条件组合出一条“最佳配方”，而是区分单晶生长、固相反应和相图筛查各自能够解决的问题，并据此设计可区分不同结构假设的实验。全文保留文献中的数值及其限制条件；结构相关体系与工艺参照体系仅用于提出实验参照。

1.1 相关体系的研究现状

Ba–Y–Si–O 四方硅酸盐系列。 早期助熔剂法生长实验将 Ba_{5+x}Y₁₃Si₈O₄₁ 报道为伴生晶相；随后的 151 组 BaO–K₂O–Y₂O₃–SiO₂–MoO₃ 体系实验表明，Y₂O₃/SiO₂ 比、MoO₃ 用量和冷却速率共同影响硅酸盐的结构类型与晶体质量 [25, 42]。现代单晶研究将相关名义组成重新审定为非化学计量 Ba_xY₂₆Si₁₆O_{71+x}；陶瓷实验还表明，玛瑙研磨可能引入 SiO₂，使块体样品偏离名义配比 [43, 44]。另一项相区研究结合能谱 (EDS)、连续旋转电子衍射 (cRED)、同步辐射粉末 X 射线衍射 (PXRD) 和中子衍射确认了 Ba₅Y₁₃[SiO₄]₈O_{8.5},

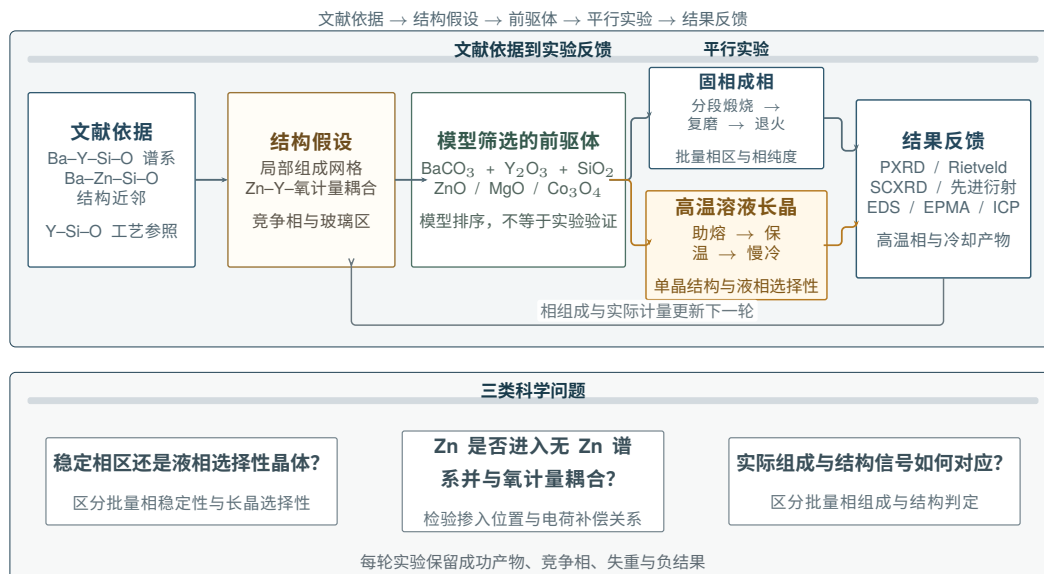


图 1: 本文的研究路线。 Ba-Y-Si-O、Ba-Zn-Si-O 和 Y-Si-O 体系先用于确定结构关系、竞争相和工艺边界; 相图与电荷平衡分析提出候选组成, 两步无机逆合成模型给出前驱体集合; 固相反应与高温溶液法晶体生长分别考察块体相区和单晶结构, 表征结果用于调整后续的成分和工艺。

综述类文献仅用于统一术语和追溯原始论文, 具体实验条件均以原始研究为准。下列资料予以排除: 仅元素或功能相近而无晶体学或合成联系的材料; 仅有计算预测而未经实验实现的相; 来源无法追溯的聚合摘要或产品信息。玻璃和微晶玻璃仅在说明晶化边界或竞争相时作为工艺参照体系, 不能据此推断目标相的结构。

记录先按 DOI 合并; 无 DOI 或存在多个版本时, 依据题名、作者、年份和来源交叉去重, 并核对作者顺序、卷页和出版状态。检索共得到 63 条候选记录, 去重和来源核验后保留 51 篇独立文献。目标相仅有一篇直接原始论文, 因此本文不宣称覆盖全面, 也不宣称已完成独立复现。

2.3 文献相关程度与条件提取

本文按研究对象与目标相的关系将文献分为五类: 目标化合物的直接报道、无 Zn 的同类 Ba-Y 四方硅酸盐、具有明确晶体学关系的结构相关化合物、可作工艺参照的合成研究, 以及不能支持结构或配方判断的非相关资料。分类针对具体论断而非整篇论文, 同一来源的结构结果和工艺信息可支持不同范围的解释。

为便于比较, 逐篇提取原料及名义配比、前处理、相形成方式、温度-时间、气氛与体系开放性、坩埚或承载体、助熔剂、冷却或生长程序以及产物与表征方法。数值仅保留在对应的原始实验记录中, 不跨研究合并。原文未明确给出的项目记为未报道, 表示本次核查未取得该信息, 并不表示原文确未报道。综合判断优先采用多篇研究在同一实验项目上的一致结果, 并分别考察单晶结构、块体相组成、实际化学计量和高温稳定性。

表 1: 目标化合物及相关 Ba–Y 硅酸盐的组成与结构依据。表中记录按文献对象与目标相的关系分类; 无 Zn 研究和结构相关研究均不构成含 Zn 目标相的独立复现。

化学式 (原文 → 本文写法)	文献	主要结构证据	与目标相的关系	结论与待研究问题
$\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$	[1]	单晶 X 射线结构鉴定 [1]	目标化合物直接文献	结论: 含 Zn 目标相及其合成路线类型; 待研究: 独立复现、块体相纯窗口及与无 Zn 系列的对应
$\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ (暂定) (本文: 保留变量 x)	[25, 42]	伴生相报道与系统相形成筛查 [25, 42]	同类 Ba–Y 四方硅酸盐	结论: 历史命名及组成筛查结果; 待研究: 精确化学计量、结构模型及与目标相的关系
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ (本文: 保留非化学计量式)	[43]	单晶非化学计量调制结构重新审定 [43]	同类 Ba–Y 四方硅酸盐	结论: 历史式的现代结构解释; 待研究: Zn 是否可进入该系列
$\text{Ba}_x\text{Y}_{26}\text{Si}_{16}\text{O}_{71+x}$ 陶瓷	[44]	块体相组成与显微结构分析 [44]	同类 Ba–Y 四方硅酸盐	结论: 非化学计量系列的块体表征需求; 待研究: 第二相、外来 SiO_2 与本征相区的区分
$\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$	[18]	EDS、cRED、同步辐射 PXRD 与中子精修 [18]	同类 Ba–Y 四方硅酸盐	结论: 独立确认的四方 Ba–Y 结构与相区参照; 待研究: 与含 Zn 目标相的位点对应
$\text{BaY}_{16}\text{Si}_4\text{O}_{33}$	[30]	单晶结构及孤立正硅酸根单元 [30]	结构相关化合物	结论: Ba–Y 正硅酸盐的结构参照; 待研究: 缺乏与含 Zn 目标相同相的依据

4 文献相关程度与可比范围

4.1 分类依据

本文按研究对象与目标相的关系组织文献, 不将整篇论文简单归入单一类别。同一研究可分别为结构、合成路线或表征方法提供依据, 每条结论仅适用于该文献直接观察的对象。文献分为五类: 目标化合物的直接报道、无 Zn 的同类 Ba–Y 四方硅酸盐、具有明确晶体学关系的结构相关化合物、可作工艺参照的合成研究, 以及不能支持目标结构或配方判断的非相关资料。分类用于说明结论的适用范围, 不评价论文质量。

目标化合物的直接报道。 化学式须与 $\text{Ba}_5\text{Y}_{12}\text{Zn}[\text{O}(\text{SiO}_4)]_8$ 一致, 且原始研究须同时给出制备过程和结构鉴定。目前仅 Ababaikei 等报道的开放体系高温溶液法生长与单晶 X 射线结构研究符合这一条件 [1]。该研究提供了唯一的直接实验起点, 但尚未建立可独立复现的块体相纯窗口。

同类 Ba–Y 四方硅酸盐。 此类研究不含 Zn, 但直接涉及四方 Ba–Y 硅酸盐的历史命名、结构重新审定或相区关系。早期工作报道了暂定的 $\text{Ba}_{5+x}\text{Y}_{13}\text{Si}_8\text{O}_{41}$ 伴生相 [25]; 现代单晶研究将相关组成重新审定为非化学计量调制结构 [43]; 独立研究则以多种衍射和成分分析方法确认了四方 $\text{Ba}_5\text{Y}_{13}[\text{SiO}_4]_8\text{O}_{8.5}$ [18]。这些结果可限定历史化学式、组成范围和竞争相, 但不能替代含 Zn 目标相的复现。

结构相关化合物、工艺参照与非相关资料。 具有可靠晶体学数据的 Ba–Zn–Si–O 或 Ba–Y–Si–O 化合物, 可用于比较多面体连接方式、阳离子位点和相变表征 [28–30, 51]; 仅提供合成路线、气氛、冷却或机械处理信息的研究, 可用于设计实验记录项目和选择表征方法 [31, 33, 41]。上述资料均不能推出目标相的组成